

# Osmania University

## REGULAR EXAMINATION ? 2024

Electronic Devices and Circuits ? Regular Examination, 2024

Subject Code: EC303 Semester: 3 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### SECTION A (Unit 1: Semiconductor Physics)

1. Explain Semiconductor Physics with a neat diagram/sketch and a suitable example. [12 Marks]
2. Write short notes on Semiconductor Physics. [12 Marks]
3. Define Semiconductor Physics. Explain its significance in Electronic Devices and Circuits with a suitable example. [12 Marks]

### SECTION B (Unit 2: PN Junction Diodes)

4. How is PN Junction Diodes implemented? Give an algorithm / procedure and explain. [12 Marks]
5. Compare and contrast the key concepts involved in PN Junction Diodes. [12 Marks]
6. Explain PN Junction Diodes with a neat diagram/sketch and a suitable example. [8 Marks]

### SECTION C (Unit 3: BJT and JFET)

7. List the advantages and limitations of BJT and JFET. Explain any two in detail. [12 Marks]
8. Discuss the engineering applications of BJT and JFET. [12 Marks]
9. Define BJT and JFET. Explain its significance in Electronic Devices and Circuits with a suitable example. [12 Marks]

### SECTION D (Unit 4: MOSFETs and Power Devices)

10. Describe the working principle of MOSFETs and Power Devices with necessary equations. [12 Marks]
11. Compare and contrast the key concepts involved in MOSFETs and Power Devices. [8 Marks]
12. How is MOSFETs and Power Devices implemented? Give an algorithm / procedure and explain. [12 Marks]

### SECTION E (Unit 5: Optoelectronic Devices)

13. Define Optoelectronic Devices. Explain its significance in Electronic Devices and Circuits with a suitable example. [12 Marks]
14. Explain Optoelectronic Devices with a neat diagram/sketch and a suitable example.

**[12 Marks]**

15. Describe the working principle of Optoelectronic Devices with necessary equations.

[12 Marks]

\*\*\* End of Question Paper \*\*\*